GCSE

Rewarding Learning

## Chief Examiner's and Principal Moderator's Report

# Digital Technology



Summer Series 2026

# Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2026 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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# GCSE DIGITAL TECHNOLOGY

# Chief Examiner's Report

## Subject Overview

Candidates generally appeared well-prepared for the examination series, reflecting the high standard of teaching and learning taking place across centres. Scripts demonstrated a solid foundational grasp of the core concepts underpinning digital systems, data handling, and technical infrastructure.

Across both the Multimedia (Route A) and Programming (Route B) pathways, the examination papers featured a comprehensive range of question formats. These included multiple-choice questions, structured technical diagrams, and extended prose items designed to differentiate across the full spectrum of candidate abilities. Papers effectively balanced accessible, recall-focused tasks with rigorous, higher-order discriminatory questions that successfully challenged the top-performing candidates.

Across all three units, most candidates correctly utilised core technical vocabulary and the more able candidates engaged well with practical application scenarios. In questions focusing on user requirements and evaluation, a significant portion of candidates confidently identified appropriate methods to adapt or improve digital systems.

A recurring issue was the inability to secure full marks on two-mark questions. While candidates frequently stated a correct initial point (e.g. an adjustment to a user interface or an evaluation benchmark), they often failed to provide the necessary secondary detail, justification, or context to gain the subsequent marks.

In sections testing more complex concepts, some candidates relied on rote-learned definitions instead of tailored application. For instance, when evaluating user requirements, candidates frequently omitted critical contextual details – such as specifying precise stakeholder engagement. Candidates should practise adapting their technical knowledge directly to the specific case studies and scenarios provided in the exam prompts.

For candidates sitting the Unit 4 paper, subtle inaccuracies in logic translation, flowchart annotations, or the completion of pseudocode sometimes resulted in preventable mark loss. Candidates would benefit from regular practice dry-running algorithms, tracing logic paths, and accurately annotating flowcharts to eliminate these errors.

# Assessment Unit 1 Compulsory Core

## Unit Overview

Overall, the paper gave candidates across the full range of abilities a good opportunity to demonstrate their knowledge and understanding of Digital Technology. There was good coverage of the specification, with an appropriate balance of question types and levels of demand. The language used was accessible and appropriate for GCSE candidates, and the scenarios provided clear and realistic contexts in which candidates could apply their knowledge.

Candidate performance was generally in line with expectations, with most questions allowing candidates to access the available marks. The combination of shorter knowledge-based questions and questions requiring candidates to apply their understanding to specific scenarios provided effective differentiation across the ability range.

The mark scheme supported fair and consistent marking, while allowing credit for alternative valid responses where appropriate. Feedback from markers was generally positive. They considered that the paper provided good coverage of the specification and that the range of questions allowed candidates of different abilities to demonstrate what they knew and understood. The use of familiar and realistic contexts was also viewed positively, particularly in helping candidates apply their knowledge rather than simply recall facts.

Overall, the paper was considered to provide an appropriate level of challenge while remaining accessible to candidates across the full ability range.

**Q1** This question was made up of Parts (a) to (h), with each part containing four short statements from which candidates were required to select the correct response. The questions were generally well answered, providing candidates with a clear opportunity to gain credit while also differentiating across the ability range. The multiple – choice questions were especially well answered, with most candidates able to identify the correct responses. Some parts were more challenging than others, but this provided an appropriate level of demand. No significant issues with the question were identified.

**Q2** **(a)** Very few candidates achieved the full two marks. A significant number of candidates gave a definition of bit rate rather than the specific response required by the mark scheme. Most candidates were able to gain one mark, showing they had some understanding of the concept, but fewer candidates demonstrated the precise knowledge needed to gain both marks.

**(b)** Candidates generally demonstrated some understanding of analogue-to-digital conversion, with most gaining at least one mark. However, relatively few candidates achieved all three marks, as this required three distinct valid points. The question provided an appropriate level of challenge and differentiated well between candidates.

**Q3** **(a)** & **(b)** Were designed to give candidates a straightforward opportunity to gain marks by expanding the acronyms PDF and RTF. The question was generally well answered. However, some candidates did not gain marks because they used ‘file’ instead of ‘format’ in their responses. This highlights the need for candidates to use the correct terminology when expanding acronyms.

**Q4** Question 4 comprised Parts (a) to (f) and covered a range of database, Big Data and data analytics concepts. Parts (a) to (d) were well answered, giving candidates good opportunities to demonstrate their knowledge and understanding.

**(a)** This part was well answered, with most candidates correctly identifying the primary key in the tblMember table, presenting little difficulty for most candidates.

**(b)** Most candidates demonstrated a sound understanding of appropriate data types, with the majority achieving at least two of the three available marks.

**(c)** Responses were generally of a good standard, although a significant number of candidates were unable to identify the correct relationship between the Members and Bookings tables. This provided useful differentiation between candidates.

**(d)** **(i)** This part was well answered; candidates correctly identified the database feature required to retrieve information.

**(ii)** Candidates generally responded well to this part. The most common incorrect response was 'form', rather than the feature used to present the information retrieved.

**(e)** **(i)** This part proved more challenging. Many responses referred to the size of data rather than the large amount of data associated with Big Data. Some candidates were able to access the second mark by identifying relevant characteristics of Big Data, including the three Vs. Overall, candidates found this concept challenging.

**(ii)** Few candidates achieved the full two marks. While many recognised that data analytics can be used to identify patterns and trends, relatively few explained how this is achieved through the application of algorithms, programmes or software. Some candidates appeared to confuse data analytics with data mining.

Question 4 was generally well answered in Parts (a) to (d), with candidates demonstrating sound knowledge of the database concepts assessed. Parts (e)(i) and (e)(ii), particularly the explanation of Big Data and data analytics, proved considerably more challenging and provided effective differentiation between candidates.

**Q5** Question 5 consisted of four parts and used a variety of question styles, including completion, naming, identification and selection. Overall, the question was very well answered and provided good opportunities for candidates across the ability range to gain credit.

**(a)** This part was very well answered. The use of a selection of supplied words enabled candidates to demonstrate their understanding of the concepts assessed, while still providing an appropriate level of challenge.

**(b)** This part was generally well answered, although performance was somewhat more varied. Most candidates were able to identify the appropriate powerful computer used as a central hub within a network.

**(c)** This part was well answered, with most candidates successfully identifying two appropriate network protocols.

**(d)** This part was also well handled. Most candidates were able to identify the correct characteristics of a ring network, with the majority achieving two of the three available marks. The selection format made the question accessible while still allowing candidates to demonstrate their knowledge.

Overall, Question 5 worked well. The variety of questioning styles gave candidates different ways to demonstrate their knowledge and provided differentiation across the ability range.

**Q6** Question 6 consisted of seven parts covering a range of spreadsheet skills and theoretical knowledge. Parts (a) to (d) were generally well answered, while Parts (e) to (g) proved more challenging. The responses highlighted some areas where candidates need to develop their understanding and application of spreadsheet concepts.

**(a)** This part was well answered. Most candidates were able to use the supplied words to complete the paragraph and identify the appropriate statements.

**(b)** Some confusion was evident between spreadsheet validation and type checking. Several candidates selected type check rather than the validation feature required by the question.

**(c)** This part was generally well answered. However, some candidates used the SUM function when a simple multiplication formula was required. This suggests that some candidates are relying on familiar functions rather than considering which formula is most appropriate for the calculation.

**(d)** This part proved more challenging than expected. A significant number of candidates identified ‘copy and paste’ rather than the appropriate spreadsheet feature for replication.

**(e)** This part was generally well answered, although some candidates demonstrated a lack of confidence when constructing or identifying appropriate spreadsheet formulae. Careless use of functions, particularly the unnecessary use of SUM, was evident in several responses.

**(f)** This part proved difficult for many candidates. A common incorrect response was to describe data modelling as the visual representation of data using charts or graphs. Candidates needed to demonstrate an understanding of what data modelling involves rather than simply describing how data can be presented.

**(g)** Responses highlighted some weaknesses in accurately identifying spreadsheet cell ranges. Many candidates failed to include the appropriate cell ranges used to create the graph, with responses such as E4:E8 and G4:G8 being common. Greater care needs to be taken when reading and interpreting spreadsheet cell ranges.

Overall, Question 6 provided good coverage of spreadsheet skills and theory. Parts (a) to (d) were generally well answered, while Parts (e) to (g) provided effective differentiation. The responses highlighted areas for further development particularly in the selection and use of formulae, data modelling and the accurate identification of cell ranges.

**Q7** Question 7 consisted of Parts (a) to (c)(ii), covering candidates' knowledge and understanding of CPU characteristics, clock speed and types of memory. Overall, the question was generally well answered and gave candidates good opportunities to demonstrate their knowledge.

**(a)** This part was well answered, with most candidates gaining at least two of the three available marks. The selection format provided a straightforward opportunity for candidates to demonstrate their knowledge of CPU characteristics.

**(b)** This part was generally well answered, although some candidates found it more difficult to achieve the full two marks. Some candidates referred to improved performance without explaining how a higher clock speed would improve performance. This also presented some difficulty in applying the mark scheme consistently where candidates had referred only to improved performance.

**(c)** **(i)** Most candidates answered this part correctly identifying ROM as the type of non-volatile memory required.

**(ii)** This part was also well answered, with most candidates correctly identifying RAM as the type of volatile memory required.

Overall, Question 7 was generally answered well and provided appropriate opportunities for candidates across the ability range to demonstrate their knowledge, with Part (b) providing the greatest opportunity for differentiation.

**Q8** Question 8 was the Quality of Written Communication question and required candidates to describe the advantages of e-commerce under two headings: **changes in shopping behaviour** and customer interaction. Overall, the question proved challenging, with relatively few candidates achieving the higher levels of the mark scheme.

**(a)** Candidates generally showed a sound understanding of changes in shopping behaviour. Many were able to identify advantages such as convenience and the ability to shop online, however, some responses lacked sufficient development to access the higher marks.

**(b)** This part proved more challenging. There was some confusion over what was meant by 'customer interaction'. Many candidates focused on the reduced need for face-to-face contact with shop staff and other customers. While this could be seen as a change in social interaction, it did not address the advantages of customer interaction in an e-commerce context. As a result, relatively few candidates were able to reach the higher levels of the mark scheme.

Overall, Question 8 was challenging. While most had a reasonable understanding of the changes in shopping behaviour, the customer interaction element was frequently misunderstood. The question provided useful differentiation and highlighted an area of the specification where candidates' understanding requires further development.

**Q9** Question 9 consisted of three parts assessing candidates' knowledge and understanding of health and safety in the workplace. Overall, the question was well answered, although some candidates did not provide sufficiently specific responses to meet the requirements of the mark scheme.

**(a)** This part was well answered, with most candidates able to identify appropriate responsibilities of an employer relating to employee health and safety. The selection format provided an accessible opportunity for candidates to demonstrate their knowledge.

**(b)** Most candidates understood the health issues that can be reduced by taking regular breaks. However, some candidates lost a mark by using the term 'back pain' instead of the more specific term 'back strain'. This highlights the importance of candidates using accurate technical terminology in their responses.

**(c)** Candidates generally identified appropriate features of a well-designed workstation. However, some responses were too general and did not clearly identify specific workstation features. Candidates need to take care to give the level of detail required by the question.

Overall, Question 9 was generally well answered. The responses showed that candidates have a good understanding of workplace health and safety, although greater care is needed when reading the question and selecting precise technical terms in written responses.

**Q10** Question 10 was worth ten marks and required candidates to match ten definitions with the appropriate Digital Technology terms. The question was very well answered, with most candidates achieving between eight and ten marks.

Most candidates showed a strong understanding of the terminology being assessed. The main area of confusion identified was between the functions of cache memory and registers.

Overall, the question worked well and provided an accessible opportunity for candidates to demonstrate their knowledge and achieve a high proportion of the available marks.

**Q11** Question 11 consisted of four parts assessing candidates' knowledge and understanding of processing methods, operating systems and data backup. Overall, the question was well answered, with Parts (a) and (c) particularly well handled.

**(a)** This part was very well answered, with very high levels of attainment observed. Most candidates successfully matched the statements to the correct processing method, and many achieved full marks.

**(b)** Candidates generally showed a sound understanding of the functions of an operating system and were able to identify appropriate resources managed by an operating system.

**(c)** This part was answered very well overall. The use of supplied terminology made the question accessible and allowed candidates an opportunity to demonstrate their knowledge.

**(d)** This part proved more challenging. While most candidates understood the reasons for backing up computer data, many found it difficult to gain the second mark. Candidates needed to apply their knowledge and give sufficiently developed reasons, and some responses lacked the detail needed for full credit.

Overall, Question 11 was well answered and provided a good balance of straightforward knowledge-based questions and opportunities for candidates to demonstrate a more developed understanding of the topics assessed.

# Assessment Unit 2 Digital Authoring Concepts

## Unit 2 Overview

This paper contained ten multi-part questions. Marks available, across individual question elements, ranged from one to a maximum of eight marks. Both questions where eight marks were available were suitably structured to reduce the need for extended writing. The questions focused instead on knowledge recall and application of knowledge in two different topic areas: the features of a GUI and how these same features could support non-expert users and then later in the paper candidates were required to complete a scenario-based test plan.

Overall, the paper was found to be accessible and allowed candidates to demonstrate a broad range of knowledge and understanding. Most candidates responded well to the questions, with many showing secure understanding of the key concepts assessed. The range of question types, including acronym expansion, gap-fill and matching tasks, allowed candidates to access marks throughout the paper with confidence. Extended response questions provided opportunities for candidates to demonstrate depth of understanding through the accurate use of subject-specific terminology. However, examiners reported that responses at times lacked precise technical vocabulary, which limited access to higher mark bands where extended responses are necessary, for example, in the Quality of Written Communication question which assessed candidate knowledge relating to the disadvantages of graphic user interfaces. It is worth noting for future examination series that extended writing questions require candidates to use the correct digital technology terminology. Candidates making use of generic terms will be unable to score in the top band. Candidate performance in the main was positive, however, with the majority of candidates able to gain marks across all sections of the paper.

Most candidates showed an increased understanding of programming structures with many obtaining full marks in questions relating to this topic area. There is a clear link between candidate understanding of assessed topics which were derived from theoretical concepts relating to controlled assessment tasks with a significant number of candidates showing a detailed understanding of query structure, relational database theory and HTML.

Centres should remind candidates to read each question carefully and respond to all task requirements, as some candidates failed to complete question parts requiring them to circle the key word needed to complete a technical phrase, resulting in avoidable loss of marks.

**Q1** The multiple-choice questions at the beginning of the paper were generally well answered. However, some misconceptions were evident. A number of candidates incorrectly believed that the  tag centres a paragraph on a web page. In addition, many candidates were unfamiliar with the HTML tags used to create a bullet list, with a significant number incorrectly selecting <bl></bl> as the correct response.

**Q2** Question 2(a) and 2(b) were answered well by many candidates. Most were able to define the term prototype accurately. However, some candidates provided examples of different types of prototypes rather than defining the term itself. In Question 2(b), some responses were too vague or generic, with candidates describing examples of user requirements rather than identifying features of a user requirements document.

In Question 2(c), most candidates demonstrated a clear understanding of the purpose of a movie timeline and a navigation structure diagram. However, some confusion was evident regarding the purpose of a data dictionary, with some candidates incorrectly identifying it as the design tool used to show the contents of footers and data within a database report. Despite the clear distinction between the terms wireframe and storyboard in the specification, some candidates confused the two and used them interchangeably when completing the closed-prose statements relating to design tools for multimedia applications and database interfaces.

**Q3** Question 3(a) highlighted gaps in candidates' use of technical terminology, particularly when describing graphical user interface (GUI) features and explaining how they benefit non-expert users. While some candidates identified relevant features, responses often lacked sufficient development, limiting access to the higher mark bands.

Questions 3(c)–(e), which focused on the range of user interfaces encountered in everyday use, were generally answered well. However, some candidates provided overly generic examples and vague explanations of advantages, particularly for natural language interfaces. These responses often relied on personal experience rather than the application of technical knowledge and subject-specific terminology. Centres should encourage candidates to develop their use of precise technical vocabulary when constructing extended responses.

**Q4** In Question 4(a)(i), most candidates were awarded one of the two available marks when defining the term plugin, indicating a need for greater focus on the precise definitions of technical terms identified in the specification. Similarly, fewer candidates were able to provide two distinct advantages of PDF files in Question 4(a)(ii). A significant number repeated the same advantage using different wording, for example, stating that PDF files are compressed and then that they have a smaller file size. Centres should emphasise the importance of providing distinct responses where multiple examples or advantages are required.

It was pleasing to see that the majority of candidates correctly identified the compressed file types in the stimulus material for Question 4(a)(iii). Likewise, most candidates successfully identified appropriate methods of ensuring an application is cross-platform compatible in Question 4(a)(iv).

In their responses to Question 4(b), candidates also performed well when identifying methods of improving accessibility in multimedia applications, often drawing on their experience of developing websites for the non-exam assessment. However, greater use of precise technical terminology is required. Responses such as 'use a zoom tool' or 'make the text bigger' were considered as references to operating system functionality or too vague as opposed to being accessibility features incorporated into an application. Candidates should instead refer to application features such as increase/decrease text size controls or text-to-speech functionality.

**Q5** Question 5 was accessible to most candidates, with many successfully identifying appropriate interactive features of e-commerce and social media applications in Question 5(a). In Question 5(b), most candidates were also able to identify interactive features of gaming applications. However, some responses focused on online communication rather than the gaming context specified in the question. Candidates who identified features specific to gaming applications were generally able to provide clear explanations of how these features enhanced gameplay.

To improve performance in similar questions, centres should encourage candidates to focus closely on the application identified in the question stem, rather than providing generic features or tools that could apply to a wide range of applications.

**Q6** Question 6(a) indicated that some candidates remain unfamiliar with the correct expansion of the acronym HTML. A number of candidates also left this question unanswered, suggesting that the question instructions had not been read carefully. Centres should emphasise the importance of reading all question instructions thoroughly as part of examination preparation.

Responses to Question 6(b) demonstrated most candidates held a secure understanding of the multimedia terms included in the closed-prose exercise, although some candidates confused the definitions of hypertext and web browser. It was pleasing to see that a significant number of candidates correctly identified the definition of event programming, which is a technical term they have often had difficulty with in previous examination sessions.

Questions 6(c)(i) and 6(c)(ii), which assessed candidates' ability to interpret HTML statements, were answered well. While candidates generally demonstrated a sound understanding of the effect of HTML tags on the content they enclose, responses to Questions 6(d)(i) and 6(d)(ii) showed they were less confident when describing the purpose of HTML tags using precise technical terminology.

Responses to Questions 6(e)(i)–(e)(iii) indicated an improvement in candidate ability to identify programming constructs from the descriptions provided.

**Q7** Question 7 provided a greater level of challenge for candidates. While Question 7(a), which assessed knowledge of relational database terminology, was generally well answered, later question parts, which required the construction of query criteria and SQL statements, proved more demanding. Common errors in candidate responses were linked to incorrect syntax, use of incorrect or incomplete table names and field headings, and the misuse of logical operators. Centres should remind candidates to use the table names and field headings provided in the question stimulus when constructing queries and SQL statements.

In Question 7(b), most candidates correctly identified the Boolean data type within the database structure provided. However, fewer candidates were able to identify a foreign key correctly. Centres should ensure candidates develop a secure understanding of more complex database concepts, particularly those applied within the non-exam assessment; this could be achieved through an increased focus on such concepts when completing controlled assessment tasks.

Most candidates correctly expanded the acronym of ERD in Question 7(c). However, a minority continue to give incorrect expansions, including terms such as Entry, Relational, or Database.

Questions 7(d), 7(e) and 7(f) assessed the construction of query criteria rather than the use of a QBE interface. Scaffolding was provided through worked examples, enabling most candidates to answer Parts (d) and (e) successfully. However, Question 7(f), which required the use of a logical operator to construct more complex query criteria, proved more challenging, with some candidates confusing the use of the AND and OR operators.

In Question 7(g)(i), most candidates provided a partially correct definition of the term macro. Many were also able to apply their knowledge successfully in Question 7(g)(ii), commonly referring to the use of macros to automate navigation buttons within form interfaces.

The SQL statement in Question 7(h) proved challenging for some candidates, with a proportion demonstrating limited understanding of the syntax required for a valid SELECT statement.

**Q8** Question 8(a) was generally well answered. However, some candidates struggled to identify the correct test data types, particularly distinguishing between correct and extreme test data, with extreme being the most commonly missed response. Some candidates also repeated the term null instead of providing appropriate null test data, while others left the relevant cell blank rather than indicating that no data should be entered (for example, by writing leave cell blank).

Question 8(b) was also answered well overall. However, some candidates provided descriptions of testing processes that lacked sufficient detail to access all available marks. Centres should remind candidates to consider the mark allocation for each question part, as this provides a useful indication of the level of detail expected in their responses.

**Q9** Question 9(a) showed mixed performance. While most candidates demonstrated a secure understanding of some of the testing methods referenced in the question, others confused key terminology, particularly alpha, beta, and white-box testing. Where candidates described A/B and beta testing, responses at times lacked sufficient detail to access all available marks. Centres should continue to develop candidates' knowledge and understanding of the full range of testing methods identified in the specification, ensuring they are able to accurately define, distinguish between, and apply the correct technical terminology associated with each method.

Question 9(b) was generally well answered, although some candidates provided basic or incomplete explanations of iterative testing. Candidates would benefit from developing a deeper understanding of the purpose and application of iterative testing rather than providing brief descriptions.

**Q10** Question 10 was generally well attempted across the cohort. Strong responses clearly linked evaluation activities to documentation such as user requirements and testing evidence. In Question 10(a)(ii) and Question 10(a)(iii), candidates had greater difficulty linking the use of design documents to the evaluation process when responding to Question 10(a)(i). Where responses were not awarded full marks in these question parts, candidates described the content or overall purpose of the document as opposed to linking to the evaluation process as required in the question stem. Part (b) was straightforward and well answered by most candidates showing candidates have a clear understanding of the stakeholders who play a role in evaluating software applications.

# Principal Moderator Report

# Assessment Unit 3 Digital Authoring Practice

## Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks via digital submission. Centres **must** ensure candidates have been provided with the correct scenario for Unit 3. This changes on an annual basis, released in early June. This information is clearly provided on the front cover of the controlled assessment task. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised. Centres may allow pupils to practise with legacy tasks and data files. Centres are encouraged, however, to demonstrate good practice through legacy tasks and provide opportunities for candidates to complete practice tasks.

## Administration

The following procedures **must** be adhered to:

- All candidates of the requested sample, including the **top** and **bottom candidate** of the sample, must be included.
- There is no requirement to copy and paste the **mark band information** into the eCandidate Record Sheet. Brief, factual comments regarding how marking was allocated are sufficient. Only **essential files** should be submitted for moderation, preferably as one **Portable Document Format**. Duplicate word processing files should not be submitted.

## Accuracy of Marking

The accuracy of assessment judgements from centres was generally high and demonstrated a sound understanding of the controlled assessment requirements. A small number of centres marked leniently and marking was adjusted. In some cases, marking was slightly harsh at the lower end of the mark range. Centres should use the 'best fit' mark band approach. The use of templates is prohibited. Candidates are encouraged to produce individual solutions. This is a controlled assessment regulation. Assessment of testing and evaluative sections were marked leniently by some centres.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that **all** centres complete internal standardisation effectively before submitting marks to CCEA. It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series.

## Task 1 Designing the multimedia and database solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- The majority of storyboards were computer generated. Rather than the focus on the volume of storyboarding, candidates should focus on the quality and detail of storyboards, for example, the design of the website could include the design of the masterpage. Database design may include a more text/tabular form/query/report design, to reduce the duplication of work.
- One example of the planning for each query, form and report is required. More detailed planning on a masterpage would reduce workload, yet still produce a high quality multimedia solution.
- Storyboards must show the sources of information and file types to be used. Some candidates did not provide the level of detail required – image sources, feedback and prototyping missing from some candidates, some plans were missing sufficient detail for third party implementation. A recurring issue in website storyboards was the omission of image file types.
- The design of professional reports is to be commended, with appropriate titles and the company logo applied appropriately.
- Some candidates failed to plan complex queries (two or more criteria) or appropriate reports (grouping, sorting and calculations).
- Some candidates did not attempt prototyping and the use of feedback in refining the solution. This section was marked leniently by some centres. 'Marking up' of prototype amendments and feedback could be applied to storyboards in a different colour.
- Some centres produced excessive detail in their design sections e.g. full prototype designs, followed by full storyboards.
- Centres should avoid creating duplicate versions of storyboards to demonstrate prototyping. Centres should encourage candidates to create a prototype and then include annotated screenshots of the final web pages, highlighting amendments from the original design. Centres are to be commended on this approach to prototyping.

## Task 2 Building the multimedia and database solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. The following points should be noted:

- If online web editors are used, e.g. Google Sites, the website URL must be presented for moderation with no additional credentials required to access the site, including access to a C2k account. When using Google Sites with a C2K account it is important to ensure that all content is set to shared access including the pages, the form and all other elements stored in Google Drive.
- The quality of website development was particularly impressive, with a noticeable increase in the number of candidates successfully using Google Sites to produce well-structured and professionally presented solutions.
- Graphics, animation and videos produced for the candidate websites were generally of a high standard.
- Overall, the standard of web authoring was high. It is positive to note many candidates using a masterpage and cascading style sheets.
- Scripted elements included: form validation, slideshow, animation, login, CSS, colour (hex).
- Accessibility elements included: browser settings, plugins (downloads), speech, magnification, alt tags.
- Database solutions incorporated appropriate relationships, user-friendly and intuitive switchboards, in addition to appropriate validation and input masks.
- Database solutions were implemented from the design document, however, the depth of complex queries and reports (grouping, sorting and calculations) was lacking, particularly grouping and calculations. Some centres included an excessive number of basic queries and reports.
- A significant number of candidates submitted queries and reports that still relied on user prompts for criteria such as instructor names or other key fields, resulting in a less efficient solution. In addition, many candidates based three reports on three queries, duplicating the requirement for both, rather than generating reports directly from tables or additional appropriately designed queries, which was not fully in keeping with the requirements of the task. A small number of centres also included date criteria within queries that had become outdated by the time of moderation, resulting in reports that produced no records.

## Task 3 Testing the database solution

- This section was marked leniently by some centres.
- Testing must include a range of valid, invalid and extreme data.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- Corrective action was not completed by a number of centres.
- ***No screenshots are required for testing; screenshots are only required to demonstrate corrective action***.

## Task 4 Evaluating the multimedia and database solution

- This section was marked leniently by some centres.
- Many candidates described their solution, rather than using evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met, fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always considered in sufficient depth.
- Evaluations remain an area for development. While most candidates completed the evaluation section, responses were often descriptive rather than analytical, with limited reflection on the effectiveness of the final solution or consideration of potential improvements.

**All documentation should be submitted as one PDF.**

# Assessment Unit 4 Digital Development Concepts

## Unit Overview

*The Unit 4 assessment paper performed well as an assessment instrument, offering an appropriate level of challenge that successfully differentiated candidate achievement across the entire ability range. The examination included a range of recall questions in the form of multiple-choice, fill-in-the-blanks, and matching terms with open-response tasks requiring computational thinking, logic, and problem-solving. This allowed less able candidates to secure marks on shorter, knowledge-based questions, while more able candidates had the opportunity to develop and complete more complex solutions which demonstrated their technical skills. Centres prepared candidates well for the examination, many displaying a sound understanding of the specification content.*

*In some cases, candidates did not always customise their answer to the question asked but provided definitions of some of the key terms in the question instead of a context-based response. Candidates should be encouraged to read the question carefully and consider the context. This was particularly evident in question 6 where the Quality of Written Communication (QWC) was assessed.*

*The Quality of Written Communication (QWC) question, relating to arrays, proved challenging for candidates. Generally speaking, candidates demonstrated some understanding of array structures but a majority provided only a basic response to this question.*

*In questions relating to data representation, such as binary arithmetic, candidates should be reminded that marks are for the method and the answer. They should be encouraged to show all stages of working when completing calculations. This is asked for in the question.*

*Algorithm-based questions indicate that some candidates found difficulty when moving from a problem specification to a solution. It is important that they practise the development and tracing of algorithms and flowcharts.*

**Q1** This question was particularly well attempted by candidates, with many achieving the full six marks available.

**Q2** *This question related to tracing algorithms.*

Parts 2(a) and 2(b) of this question were well handled by candidates. However, candidates found Part 2(c) more challenging. A significant number of candidates did not select the correct response. This question presented a nested for-loop with an embedded if-statement. Responses suggest that some candidates had difficulty tracing and interpreting the structure. The question successfully discriminated between candidates and those with more developed logical thinking skills achieved the mark.

**Q3** **(a)** This question was very well answered by candidates. A significant majority of candidates achieved the full four marks showing a good understanding of user-defined functions and their properties.

**(b)** **(i)** This question was designed to assess candidates' knowledge of string functions. Again, the question was well attempted by the majority of candidates and many full mark responses were observed. A common incorrect response given by candidates was the use of 'substring' instead of 'concatenate'.

**(ii)** This question required candidates to complete an algorithm to find the average of five temperatures. A significant number of candidates achieved full marks in this part of the question. Candidates lost marks, particularly in the final two responses because they did not refer to the variable Message or used an output statement instead of an assignment statement. Again, this is a case where candidates did not read the stem of the question. The requirement of the algorithm is specified in the stem and some of the algorithm has been completed to guide candidates in their response.

**(c)** **(i)** Candidates were required to explain the term syntax error when it occurs in a program. In order to achieve full marks candidates should have described what a syntax error is and provide an example. Many candidates provided good examples, however, some were unable to accurately give a description or definition of a syntax error. The command word 'Explains', used in the stem of the question, should prompt candidates to provide an extended answer beyond a definition, which illustrates their understanding.

**(ii)** Few candidates achieved full marks in this question related to auto indentation and code completion. The question was asked in the context of how these features could help a programmer. So, responses were expected to refer to "improving readability", "assisting debugging", "reducing errors" etc. Whilst the majority of candidates were able to provide some of these responses many defaulted to restating the stem. For example, for auto indentation a common response was "it indents the code automatically" with no reference to how this helps the programmer. Again, this question differentiated between candidates across the range. Less able candidates find it difficult to articulate how the features assisted in program development.

**Q4** As always, performance in the binary arithmetic question was very good indicating great preparation of candidates by their teachers. A significant majority of candidates achieved high marks in the calculation parts of this question. A minority of candidates did not circle the overflow Part (c) of the question. Another recurring issue was identified in Part (a) of the question where some candidates used 7 bits instead of 8, as directed in the question, when responding. This resulted in the loss of one mark. Candidates should note that it is necessary to show conversion work by using either division by two or illustrating place values.

**Q5** Question 5 tested candidates' knowledge about Boolean operators and their ability to construct rules using these operators.

**(a)** This was a closed response table where candidates had to identify the Boolean operator given its description. A significant number of candidates achieved full marks in this part of the question.

**(b)** Was not so well answered. Many candidates were unable to accurately construct the rule required using a description and some of the related output. Candidates commonly correctly identified the output values for column C but did not correctly construct the rule for the column. Stronger candidates were generally able to identify the required operator and complete the truth table accurately, making this a good discriminator across the ability range.

**Q6** *The **Quality of Written Communication (QWC)** was not as well answered as in previous years. Candidates seemed to struggle with the content relating to arrays. However, even some good responses did not fully address the question posed. At this level candidates are expected to know about the basic structure and function of arrays and how they can be manipulated. Candidates found difficulty in articulating this and even omitted aspects of the question in their response. A number of candidates did not address the question at all or provided incorrect responses which meant they were placed in the Level 0 band. The strongest candidates provided excellent answers demonstrating a full understanding of these concepts. This question successfully differentiated across the range of abilities and identified an area where many candidates could improve their performance.*

**Q7** This question tested candidates' knowledge of flow charts and their ability to apply a flow chart to a logic problem.

Parts (a) and (c) of the question were well attempted by the majority of candidates, many achieving full marks by correctly identifying the elements of a flow chart.

Part (b) candidates were required to construct a flow chart given a problem. They were provided with the flow chart structure. This was intended to guide them through the process of creating the solution. Some candidates included two statements in the input/output boxes. It should be noted that this was acceptable provided there were no mix of statements as each symbol should represent one type of action only. Therefore whilst two input statements placed in one box was acceptable, a mix of input and output in one box was not. Generally speaking, this question was well attempted.

**Q8** **(a)** A significant number of candidates achieved one mark in this part of the question by correctly explaining that abstraction involves filtering information. Some candidates incorrectly confused abstraction with decomposition. Some less able candidates stated that abstraction "makes code simpler" without further elaboration. This is too vague for credit. So, whilst candidates were able to describe what abstraction is many were unable to provide a satisfactory elaboration on what abstraction achieves or what information (not required for the solution) is removed by carrying out abstraction.

**(b)** This closed response question about pseudo-code was very well attempted and a significant number of students achieved four marks showing a sound understanding of its properties.

**(c)** **(i)** This question required candidates to select an appropriate validation check and whilst many correctly identified a length check for the password fewer correctly identified the format check. At this level candidates should be able to select correctly from the range of validation checks studied on the specification and used in the construction of their coursework.

**(ii)** Candidates were required to complete an algorithm. There are a number of points to be made here. Firstly, when variable names are provided in the algorithm candidates should continue their use throughout. A number of candidates chose to use other variable names which made their algorithm incorrect. It should be noted that both the username AND the password needed to be checked in order to achieve a successful login. Some candidates checked only one of these variables. Another incorrect response was the use of an algorithmic statement like "if password is correct". This was not acceptable as the values for each of the variables were provided in the stem. Algorithms containing nested if statements and single if statements for each of username and password were acceptable. Many candidates achieved full marks in this question paying attention to the detail provided in the stem of the question.

**(d)** **(i)** This question assesses a candidate's understanding of the need for permanent (non-volatile) storage within a specific context, such as a login system. To achieve full marks, candidates need to explicitly identify that data must be retained when the program is closed or between separate executions of the software. A number of candidates responded by saying that *"the data is stored because it needs to be compared when the login process is taking place."* This is a significant conceptual error. The candidate is confusing the *runtime validation process* with the r*eason for storage mechanism selection*. The comparison happens in RAM during execution, whereas the question tests why the data must reside in secondary storage long-term.

**(ii)** Most candidates were able to correctly provide two file handling functions.

**Q9** **(a)** Many candidates achieved full marks in this part of the question. However, some candidates chose incorrect data types. The details provided in the stem of this question should have pointed candidates to the correct data types for the variables; despite this, a significant number of candidates chose String and/or Integer. There is a clear tendency for candidates to default to String and Integer as catch-all answers when they are unsure or when they skip over the scenario details. Candidates should look for explicit clues in the text, for example, the single quotation marks surrounding the DogSize values 'S', 'M', 'L' and the fact that GroomCost was shown to two decimal places to avoid this pitfall.

**(b)** This was well attempted by the significant majority of candidates. Where candidates lost marks, they had not structured assignment statements correctly or instead of assigning the variable GroomCost to a value they chose to output it instead, losing a mark in each case. Again, candidates should be instructed to carefully read the requirements of the question.

**(c)** In this question candidates were required to complete an algorithm. The purpose of the algorithm was to calculate the TotalGroomCost. More able candidates achieved full marks in this question. Some recurrent issues included *assigning the value of DogCollected to "true" in the if statement, testing for a DogDistance < 5 and using the wrong variable name in the running total*. In addition, some candidates introduced new variable names which was completely unnecessary and made the algorithm incorrect. Candidates should note that where variable names are supplied in the stem of the question these should be used throughout the algorithm.

**Q10** These questions evaluate a candidate's operational understanding of standard sorting routines. Candidates were required to manually demonstrate the incremental, step-by-step stages of an insertion sort on an unsorted array and accurately complete the missing code components of a bubble sort implementation.

**(a)** Many candidates incorrectly treated the process like a bubble sort, swapping adjacent pairs across the entire array in a single pass, leading to incorrect responses in the first three stages. Ultimately many candidates achieved a mark for the last stage where the array was fully sorted.

**(b)** A high proportion of candidates ignored zero indexing and did not correctly identify the loop boundaries, creating loops that would allow the index of the array to go out of bounds so any program using the solution would crash. They also demonstrated difficulties with swapping values and did not use the Temp variable correctly. Fewer candidates achieved full marks in this part of the question.

**Q11** Question 11 consisted of three parts assessing candidates' knowledge and understanding of testing and tracing an algorithm to find errors. Overall, the question was well answered, particularly Parts (a) and (b), and allowed for candidates across the range of abilities to achieve marks.

**(c)** Required candidates to identify the line in the algorithm which was incorrect. This was well attempted with a significant majority of candidates identifying the correct line. The second part of this question required the candidates to correct the error and this was also well attempted, with many candidates achieving full marks. Where candidates lost marks, they did not construct the condition correctly or they did not use the correct inequality in the condition.

**Q12** **(a)** Performance was strong here: candidates accurately identified the target groups for system evaluation, leading to a high volume of full marks.

**(b)** **(i)** This part of the question on documented testing proved to be slightly more challenging for candidates. Many candidates provided correct responses relating to making changes or improvements to the system, but some did not follow up by providing sufficient detail for a second mark. Again, careful consideration of the context, in this case supporting evaluation, may have assisted.

**(ii)** Whilst answers here were better, some candidates still found difficulty in providing responses that attracted two marks. In these cases, only one use was provided for example, 'could be used as a checklist' but they missed the second mark by not connecting the user requirements to the broader context of system evaluation.

# Assessment Unit 5 Digital Development Practice

## Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks via digital submission. Centres **must** ensure candidates have been provided with the correct scenario for Unit 5. This changes on an annual basis, released in early June. This information is clearly provided on the front cover of the controlled assessment task. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised. Centres are encouraged however, to demonstrate good practice through legacy tasks and provide opportunities for candidates to complete practice tasks.

## Administration

The following procedures **must** be adhered to:

- **Program usernames and passwords must be recorded accurately on the e-candidate record sheet**. If this is not completed, moderation cannot continue.
- All candidates of the requested sample, including the **top** and **bottom candidate** of the sample, must be included.
- There is no requirement to copy and paste the **mark band information** into the eCandidate Record Sheet. Brief, factual comments regarding how marking was allocated are sufficient.
- Only **essential files** should be submitted for moderation. Original data files that have been converted to **Portable Document Format** should not be included.

## Accuracy of marking

The accuracy of assessment judgements from centres was sound and demonstrated a sound understanding of the controlled assessment requirements. High quality work was produced by centres and application of the mark scheme was highly accurate.

The use of templates is prohibited. Candidates are encouraged to produce individual solutions. Assessment of testing and evaluative sections continues to be an area of concern for some centres, regarding leniency.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that all centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures. It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshot reports for controlled assessment tasks. Thank you to all centres for their efforts this series.

Stronger candidate submissions showed clear planning, well-structured development, thorough testing and detailed evaluations. In some weaker candidate submissions, evidence lacked sufficient detail or clarity, particularly in relation to testing and evaluation, which limited access to the higher mark bands.

## Task 1 Design a solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- In some instances, user requirements were extracted from the task, without development or interpretation. Target audience needs must also be fully identified.
- The design component was particularly well implemented with high-quality flowcharts and pseudocode included effectively.
- Validation and error trapping techniques were designed appropriately, however, use of user feedback to refine the solution was omitted by a small number of centres and marked leniently.

## Task 2 Building a solution

Many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. Centres are to be highly commended for their approach to this task. The following points should be noted:

- **Remember, program usernames and passwords must be noted on the eCandidate record sheet.**
- Marking was highly accurate across the sampled centres and the programming solutions were of a high standard. A small number of centres applied the mark scheme leniently and did not follow the mark band approach.
- A key issue at the top mark band remains validation. Some highly marked programs crashed with basic invalid inputs, which is not consistent with top-level performance.
- Solutions of the highest standard gave due consideration to system robustness.
- All solutions must be annotated.

## Task 3 Testing the solution

- Test plans were well structured, in tabular format (showing tests, expected outcome, actual outcome and fixes).
- It is very important that testing incorporates a range of tests (black and white box testing) and includes valid, invalid and extreme test data. Extreme testing of data was lacking in a number of centres.
- Some candidates designed an excessive number of tests and included screenshots for all testing cases.
- No screenshots are required for testing; screenshots are required to demonstrate corrective action. Corrective action was not completed by a number of centres.

## Task 4 Evaluating the solution

- Some candidates described their solution, rather than using evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met, fully met, partially met or not met.
- To achieve marks within the top mark band, evaluations should move beyond description and provide detailed analysis of how effectively the solution meets the user requirements, including consideration of performance, robustness and possible refinements.
- A small number of centres marked the performance, robustness and refinements section leniently.

**All documentation should be submitted as one PDF.**

# Contact details

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